$x' = \lambda x$

$\frac{dx}{dt} = \lambda x$

$x = x(t)$, $\lambda \in \mathbb{R}$

fct nec.

t variabilă

$x' = \lambda x \Rightarrow x(t) = Ce^{\lambda t}$

$C \in \mathbb{R}$

$x : \mathbb{R} \longrightarrow \mathbb{R}$

Sisteme de ecuații diferențiale liniare cu coeficienți constanți

$$\begin{cases} x_1' = a_{11}x_1 + \dots + a_{1n}x_n + b_1(t) \\ \dots\dots\dots\dots\dots\dots \\ x_n' = a_{n1}x_1 + \dots + a_{nn}x_n + b_n(t) \end{cases}$$

$A = (a_{ij})_{i,j=\overline{1,n}} \in \mathcal{M}_n(\mathbb{R})$

$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ $\quad X = X(t)$

$b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ $\quad b = b(t)$

- t variabilă
- $x_1(t), \dots, x_n(t)$ fct nec.
- $a_{11}, \dots, a_{nn} \in \mathbb{R}$
- $b_1, \dots b_n : I \subseteq \mathbb{R} \longrightarrow \mathbb{R}$ cnt.

$\Rightarrow X' = AX + b(t)$

- $b(t) = 0$ $\forall t$ — Syst. omogen
- $b(t) \neq 0$ — Syst. neomogen

Sisteme de ec. dif. liniare cu coef. constanți omogene $X' = AX$

Thm. Problema Cauchy $\begin{cases} X' = AX \\ X(t_0) = X_0 \end{cases}$ are soluție unică $\forall t_0 \in \mathbb{R}, X_0 \in \mathbb{R}^n$

Consec. Problema Cauchy $\begin{cases} X' = AX \\ X(t_0) = 0 \in \mathbb{R}^n \end{cases}$ are doar sol. banală $X(t) = 0 \ \forall t \in \mathbb{R}$

exemplu: $\begin{cases} x' = 4x - 3y \\ y' = 2x + y \end{cases}$, $\underline{x(0) = 0}$, $\underline{y(0) = 0}$ $\Rightarrow x(t) = y(t) = 0 \ \forall t \in \mathbb{R}$

Prop. 1) $S_0 = \{ X : \mathbb{R} \longrightarrow \mathbb{R}^n \mid X \text{ derivabilă}, X' = AX \}$
mulțimea soluțiilor sist. de ec. dif. liniare omogene $X' = AX$
$\Rightarrow S_0$ spațiu vectorial / $\mathbb{R}$

2) Fie $t_0 \in \mathbb{R}$ și $X_1, \dots, X_p \in S_0$. Atunci $X_1 \dots X_p$ liniar ind.

$\Leftrightarrow$ $X_1(t_0), \dots, X_p(t_0) \in \mathbb{R}^n$ liniar independente.

3) $\dim_{\mathbb{R}} S_0 = n$

Dem. 1) $X_1, X_2 \in S_0 \Rightarrow X_1 + X_2 \in S_0$ (principiul superpoziției)

$X \in S_0, \alpha \in \mathbb{R} \Rightarrow \alpha X \in S_0$ - exercițiu

2) $\Rightarrow$" Fie $\alpha_1 \dots \alpha_p \in \mathbb{R}$ a.î. $\alpha_1 X_1(t_0) + \dots + \alpha_p X_p(t_0) = 0$.

Fie $X(t) = \alpha_1 X_1(t) + \dots + \alpha_p X_p(t)$

$X_1, \dots, X_p \in S_0$ sp. vectorial $\Rightarrow X \in S_0$

$\begin{cases} X' = AX \\ X(t_0) = 0 \end{cases}$

$\Rightarrow$ $X(t) = 0 \quad \forall t \in \mathbb{R}$

$\alpha_1 X_1(t) + \dots + \alpha_p X_p(t) = 0 \quad \forall t \in \mathbb{R}$

$X_1, \dots, X_p$ funcții liniar independente $\Rightarrow \alpha_1 = \dots = \alpha_p = 0$

$\Rightarrow$ $X_1(t_0), \dots, X_p(t_0) \in \mathbb{R}^n$ vectori liniar independenți.

$\Leftarrow$ Fie $\alpha_1 \dots \alpha_p \in \mathbb{R}$ a.î. $\alpha_1 X_1 + \dots + \alpha_p X_p = 0$

$\alpha_1 X_1(t) + \dots + \alpha_p X_p(t) = 0 \quad \forall t \in \mathbb{R}$

$t = t_0$

$\Rightarrow \alpha_1 X_1(t_0) + \dots + \alpha_p X_p(t_0) = 0 \in \mathbb{R}^n$

$X_1(t_0), \dots, X_p(t_0) \in \mathbb{R}^n$ liniar indep. $\Rightarrow \alpha_1 = \dots = \alpha_p = 0$

$\Rightarrow X_1, \dots, X_p \in S_0$ liniar indep.

3) 2) $\Rightarrow \dim_{\mathbb{R}} S_0 \leq n$

$=$

Fie $e_1 \dots e_n$ vectorii din baza canonică din $\mathbb{R}^n$

$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \dots e_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$

$\begin{cases} X' = AX \\ X(0) = e_1 \end{cases}$ pb. Cauchy are sol. unică $X_1(t) \Rightarrow X_1 \in S_0$

$\begin{cases} X' = AX \\ X(0) = e_n \end{cases}$ ——— ——— $X_n(t) \Rightarrow X_n \in S_0$ $\rightarrow$

$X_1(0) = e_1, \dots, X_n(0) = e_n$
vect. liniar indep.

2)
$\Rightarrow X_1 \dots X_n \in S_0$ fct. liniar indep. $\Rightarrow X_1 \dots X_n$ bază în S_0

$\Rightarrow \dim_{\mathbb{R}} S_0 = n$.

Def. Fie $\{X_1 \dots X_n\}$ bază în $S_0 \Rightarrow X_1 \dots X_n$ **sistem fundamental de soluții** ale sistemului de ec. dif. $X' = AX$.

Soluția generală a sist. $X' = AX$ este $X(t) = C_1 X_1(t) + \dots + C_n X_n(t)$

$C_1 \dots C_n \in \mathbb{R}$ constante reale (unice)

$$X(t) = \underbrace{(X_1(t) | \dots | X_n(t))}_{M(t)} \begin{pmatrix} C_1 \\ \vdots \\ C_n \end{pmatrix}$$

matrice fundamentală de soluții (inversabilă)

Pb: Cum determinăm un sistem fundamental de soluții pt $X' = AX$?

$n = 1$ $\quad x' = \lambda x$ $\quad x(t) = Ce^{\lambda t}$, $C \in \mathbb{R}$

$n \geq 1$ $\quad X' = AX$ căutăm o sol. de forma $X(t) = e^{\lambda t} v$, $\lambda \in \mathbb{R}$, $v \in \mathbb{R}^n$

$X'(t) = \lambda e^{\lambda t} v$

$X' = AX \Rightarrow \lambda e^{\lambda t} v = A e^{\lambda t} v \Rightarrow \lambda v = Av$

$Av = \lambda v$

Propoziție. $A \in \mathcal{M}_n(\mathbb{R})$ diagonalizabilă (peste $\mathbb{R}$)

$\lambda_1 \dots \lambda_n$ val. proprii

$v_1 \dots v_n$ vect. proprii liniar indep. coresp. val. pr. $\lambda_1 \dots \lambda_n$

$(Av_i = \lambda_i v_i, \quad i = \overline{1,n})$

$\Rightarrow X_1(t) = e^{\lambda_1 t} v_1, \dots, X_n(t) = e^{\lambda_n t} v_n$ sist. fundamental de sol. pt $X' = AX$ $\Rightarrow$ sol. gen. este

$X(t) = C_1 e^{\lambda_1 t} v_1 + \dots + C_n e^{\lambda_n t} v_n$, $C_1 \dots C_n \in \mathbb{R}$.

$n = 2$ $\quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ $\quad X = \begin{pmatrix} x \\ y \end{pmatrix}$ $\quad \begin{cases} x' = ax + by \\ y' = cx + dy \end{cases}$

$P_A(X) = X^2 - (a+d)X + (ad - bc)$

- $\lambda_1 \neq \lambda_2$, $\lambda_1, \lambda_2 \in \mathbb{R}$
- $\lambda_1 = \lambda_2 \in \mathbb{R}$
- $\lambda_1, \lambda_2 \in \mathbb{C}$, $\lambda_2 = \overline{\lambda_1}$

Cas 0 $\quad A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ matrice diagonale

$\begin{cases} x' = \lambda_1 x \\ y' = \lambda_2 y \end{cases}$ (équations sont découplées) $\quad x(t) = C_1 e^{\lambda_1 t}$, $y(t) = C_2 e^{\lambda_2 t}$, $C_1, C_2 \in \mathbb{R}$.

exemples. $\quad \begin{cases} x' = x \\ y' = 2y \end{cases} \Rightarrow \begin{cases} x(t) = C_1 e^{t} \\ y(t) = C_2 e^{2t} \end{cases}$, $C_1, C_2 \in \mathbb{R}$.

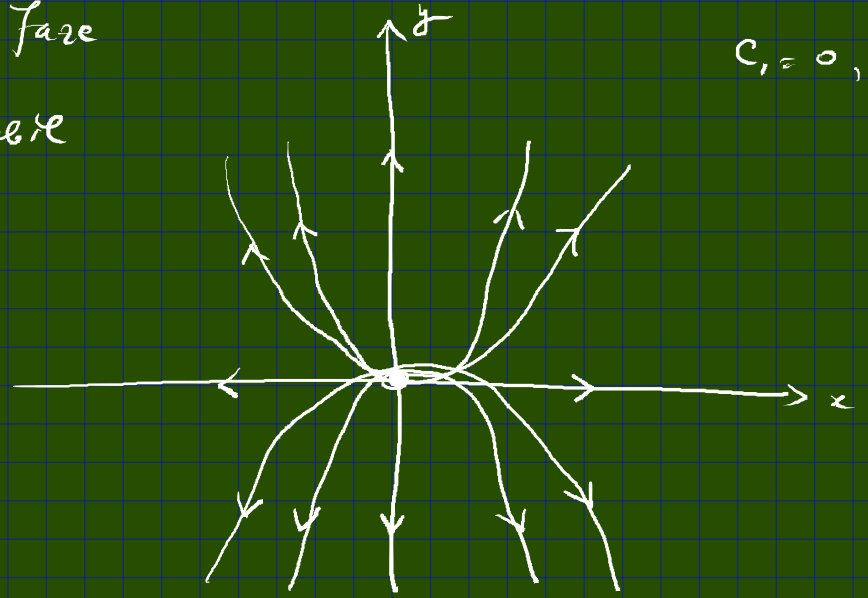
$0 < \lambda_1 < \lambda_2$

$C_1 = C_2 = 0 \Rightarrow x(t) = 0$, $y(t) = 0$ $\forall t \in \mathbb{R}$

$C_1 \neq 0$, $C_2 = 0 \Rightarrow y(t) = 0$ $\forall t \in \mathbb{R}$

Portret de faze

$(0,0)$ nod instabil



$x(t) = C_1 e^t$

$C_1 = 0,\ C_2 \neq 0 \Rightarrow x(t) = 0$

$y(t) = C_2 e^t$

$C_1 \neq 0,\ C_2 \neq 0$

$\frac{x^2}{y} = \frac{C_1^2}{C_2}$

$y = x^2 \cdot \frac{C_2}{C_1^2}$

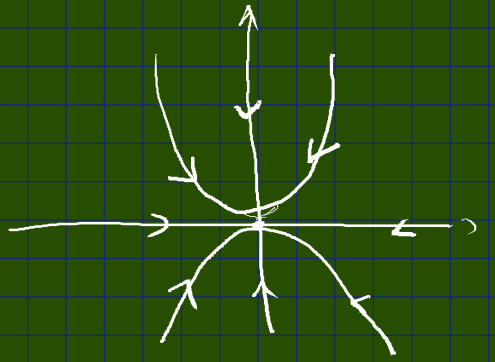
$\begin{cases} x' = -x \\ y' = -2y \end{cases}$

$\Rightarrow x(t) = C_1 e^{-t}$

$\Rightarrow y(t) = C_2 e^{-2t}$

$\lambda_2 < \lambda_1 < 0$

$(0,0)$ nod stabil atractiv



Cazul 1. A diagonalizabilă $\lambda_1 \neq \lambda_2$ $\lambda_1, \lambda_2 \in \mathbb{R}$

$\downarrow \quad \downarrow$

$v_1 \quad v_2$

$\Rightarrow$ Sol. gen. a sist. de ec. dif $\begin{cases} x' = ax + by \\ y' = cx + dy \end{cases}$ va fi

$$X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^{\lambda_1 t} v_1 + C_2 e^{\lambda_2 t} v_2 \qquad C_1, C_2 \in \mathbb{R}$$

Exemplu. ecosistem - 2 specii în competiție pentru hrană

$x(t)$, $y(t)$ t - ani

$$\begin{cases} x' = 1,6x - 0,3y \\ y' = -0,8x + 1,4y \end{cases} \qquad x(0) = y(0) = 15$$

$$A = \begin{pmatrix} 1,6 & -0,3 \\ -0,8 & 1,4 \end{pmatrix}$$

$$P_A(X) = X^2 - 3X + (2,24 - 0,24)$$

$$= X^2 - 3X + 2 \quad \begin{cases} \lambda_1 = 1 \\ \lambda_2 = 2 \end{cases}$$

$\lambda_1 = 1$ $\quad V_{\lambda_1} = \{ \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \in \mathbb{R}^2 \mid A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \lambda_1 \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \}$

$$\begin{cases} 1,6x_1 - 0,3y_1 = x_1 \\ -0,8x_1 + 1,4y_1 = y_1 \end{cases} \Rightarrow -0,3y_1 = -0,6x_1 \quad y_1 = 2x_1$$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} x_1 \\ 2x_1 \end{pmatrix} \mid x_1 \in \mathbb{R} \right\} = Sp\left\{ \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{v_1} \right\}$$

$\lambda_2 = 2$ $\quad V_{\lambda_2} = \left\{ \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \in \mathbb{R}^2 \mid A\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \lambda_2 \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right\}$

$$\begin{cases} 1,6x_2 - 0,3y_2 = 2x_2 \\ -0,8x_2 + 1,4y_2 = 2y_2 \end{cases} \Rightarrow -0,3y_2 = 0,4x_2$$

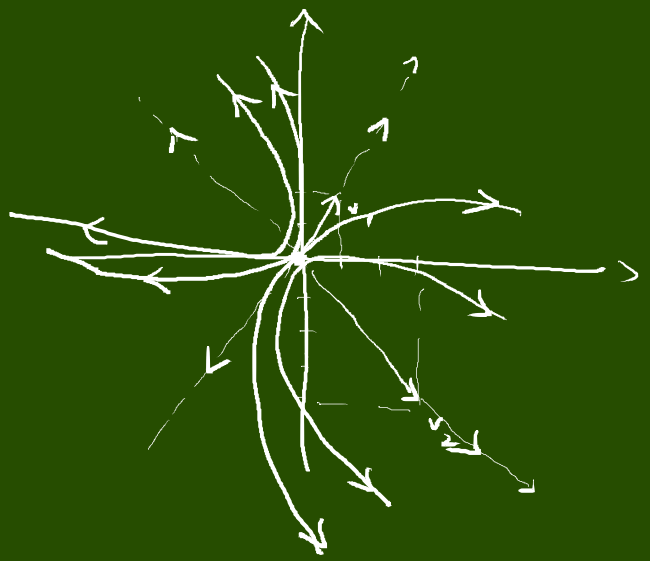
$$y_2 = -\tfrac{4}{3}x_2$$

$$V_{\lambda_2} = \left\{ \begin{pmatrix} x_2 \\ -\frac{4}{3}x_2 \end{pmatrix} \mid x_2 \in \mathbb{R} \right\} = Sp\left\{ \underbrace{\begin{pmatrix} 3 \\ -4 \end{pmatrix}}_{v_2} \right\}$$

$$\Rightarrow \quad X(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^{t}\begin{pmatrix} 1 \\ 2 \end{pmatrix} + C_2 e^{2t}\begin{pmatrix} 3 \\ -4 \end{pmatrix} \qquad C_1, C_2 \in \mathbb{R}$$

$$= \underbrace{\begin{pmatrix} 1 & 3 \\ 2 & -4 \end{pmatrix}}_{P_{B_c}^{B=\{v_1,v_2\}}} \underbrace{\begin{pmatrix} C_1 e^{t} \\ C_2 e^{2t} \end{pmatrix}}$$

$\rightarrow$ coordinate soluzioni $\begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$ in regime di base $\{v_1, v_2\}$



$$\begin{cases} x(t) = C_1 e^t + 3C_2 e^{2t} \\ y(t) = 2C_1 e^t - 4C_2 e^{2t} \end{cases}$$

Matlab

$$\begin{matrix} x(0) = 10 \\ y(0) = 10 \end{matrix} \Rightarrow \begin{cases} C_1 + 3C_2 = 10 \;/\cdot 2 \\ 2C_1 - 4C_2 = 10 \end{cases}$$

$$10C_2 = 10$$

$$C_2 = 1$$

$$C_1 = 7$$

$$\Rightarrow \begin{cases} x(t) = 7e^t + 3e^{2t} \nearrow \infty \\ y(t) = 14e^t - 4e^{2t} \xrightarrow{t \to \infty} -\infty \end{cases}$$

$t_0 \quad y(t_0) = 0$

$$14e^{t_0} = 4e^{2t_0}$$

$$\frac{7}{2} \quad \frac{14}{4}$$

$$t_0 = \ln 3.5 \geq 1$$

$$< 2$$

Cas 2 : $P_A(X) = (X-\lambda)^2$

2.1. A diagonalizable $\quad A = P D P^T = P \underbrace{\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}}_{\lambda I_2} P^T = \lambda I_2 = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$

$\longrightarrow$ cas 0

2.2. A non diagonalizable $\quad \dim_{\mathbb{R}} V_\lambda = 1$

Thm. Hamilton Cayley $\quad (A - \lambda I_2)^2 = 0_2$

Soit $v_2 \in \mathbb{R}^2$, $v_2 \notin V_\lambda$ $\Rightarrow$ $v_1 = (A - \lambda I_2) v_2$

$\begin{cases} v_1 \neq 0 \\ v_1 \in V_\lambda \end{cases}$ $\quad (v_1 = 0 \Leftrightarrow (A - \lambda I_2) v_2 = 0 \Leftrightarrow A v_2 = \lambda v_2$
$\Leftrightarrow v_2 \in V_\lambda$, faux)

$(\Leftrightarrow A v_1 = \lambda v_1 \Leftrightarrow (A - \lambda I_2) v_1 = 0$
$\Leftrightarrow \underbrace{(A - \lambda I_2)(A - \lambda I_2)}_{0} v_2 = 0$ ✓)

$\Rightarrow v_1, v_2 \in \mathbb{R}^2$ libre indépendante

$$\begin{cases} Av_1 = \lambda v_1 \\ Av_2 = (A - \lambda I_2 + \lambda I_2)v_2 = (A - \lambda I_2)v_2 + \lambda v_2 = v_1 + \lambda v_2 \end{cases}$$

$$A\underbrace{(v_1 \mid v_2)}_{P \text{ inversabilă}} = \underbrace{(v_1 \mid v_2)}_{P} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$$

$$A = P \underbrace{\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}}_{\text{matrice Jordan } J} P^{-1}$$

$X' = AX$ $\qquad X_1(t) = e^{\lambda t} \cdot v_1 \qquad (v_1 \in V_\lambda)$

$X_2(t) = e^{\lambda t}(tv_1 + v_2)$

1) $X_1, X_2 \in S_0$ soluții ale sist. de ec. diff. $X' = AX$

2) X_1, X_2 liniar indep.

1) X_1 sol. ✓

X_2 sol. $\quad X_2'(t) = \lambda e^{\lambda t}(tv_1 + v_2) + e^{\lambda t} v_1$

$\lambda e^{\lambda t}(tv_1 + v_2) + e^{\lambda t} v_1 = A e^{\lambda t}(tv_1 + v_2)$

$\lambda t v_1 + \lambda v_2 + v_1 = tAv_1 + Av_2$

$= t\lambda v_1 + v_1 + \lambda v_2$

$\Rightarrow X_2(t) = e^{\lambda t}(tv_1 + v_2) \in S_0$

2) X_1, X_2 lin. indep $\longleftrightarrow$ $X_1(0), X_2(0)$ lin indep. $\checkmark$

$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\quad v_1 \qquad v_2$

$\Rightarrow$ $X_1(t), X_2(t)$ SFS. Fundam. de sol.

$\Rightarrow$ sol. gen. $X(t) = C_1 e^{\lambda t} v_1 + C_2 e^{\lambda t}(tv_1 + v_2)$, $C_1, C_2 \in \mathbb{R}$